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Machine-learning-guided discovery of direct-gap Si-Ge alloys in hexagonal and orthorhombic silicon allotropes

Shuo Tao¹, Li Zhu^{1,*}¹Department of Physics, Rutgers University, Newark, NJ 07102, United States of America

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ABSTRACT

The pursuit of direct band gap silicon allotropes with enhanced photon absorption for photovoltaic applications has remained a longstanding challenge. While crystal structure prediction offers a theoretical route to discover such materials, it is often computationally expensive—even when aided by evolutionary algorithms like genetic algorithms or swarm intelligence. A more cost-effective alternative involves substituting silicon atoms in known stable or metastable Si phases with group IV or III-V elements such as Ge or AlP, potentially yielding direct-gap materials. In this work, we systematically investigate $\text{Si}_{1-x}\text{Ge}_x$ alloys derived from 4H-Si and $Cmcm$ - Si_{24} prototypes to identify promising candidates with direct band gaps. Leveraging our recently developed machine learning model, EOSnet, which incorporates Gaussian Overlap Matrix fingerprints to embed many-body interactions, we prescreen the candidates to eliminate structures with undesirable electronic properties. This screening step significantly reduces computational cost prior to hybrid-functional density functional theory calculations. Our study demonstrates the potential of Ge substitution to enhance light absorption in Si allotropes, laying the groundwork for future design of efficient Si-based photovoltaic materials.

1. Introduction

Silicon has long been the backbone of commercial photovoltaics due to its abundance, mature processing infrastructure, and inherent long-term stability [1]. However, diamond-structure silicon (d -Si) suffers from an indirect band gap of ~ 1.1 eV (with a much larger direct gap of ~ 3.4 eV) [2,3], which severely limits its optical absorption. Consequently, decades of intensive engineering on Si-based solar cells [4–15] have pushed their efficiencies to within a few percentage points below the Shockley-Queisser limit for single-junction devices [16], leaving little room for further improvement. To surpass this plateau, new semiconductor materials are needed that retain practical advantages of Si but exhibit direct band gaps in the ideal range of 1.1–1.4 eV for optimal solar energy harvesting [17].

One promising avenue is to explore silicon allotropes (alternative crystal forms of Si) that possess improved optoelectronic properties. High-pressure phases of Si recovered at ambient conditions have been discovered — for example, the recently synthesized 4H-Si [18] (hexagonal polytype) and $Cmcm$ - Si_{24} [7] (a clathrate-like orthorhombic structure) — which exhibit quasi-direct or near-optimal band gaps. The novelty of these phases lies in their distinct bonding and electronic structures compared to d -Si. However, to date none of the known allotropes of Si fulfills the direct band gap criterion in the 1.1–1.4 eV range. For instance, commonly studied metastable phases like R8-Si,

BC8-Si, or lonsdaleite Si have either indirect gaps (0.24–0.95 eV) or direct gaps well below the optimal range [19–21]. Even computational searches for purely silicon-based direct-gap crystals (e.g., via crystal structure prediction [22–25]) have had limited success — some hypothetical Si allotropes with direct gaps were predicted, but often their bandgap falls outside the ideal range or they face synthesis challenges [5,6,26–34]. This impasse motivates alloying strategies as an alternative approach.

Alloying silicon with isovalent elements or compound substitutions can modulate the band structure and potentially induce direct-gap behavior while leveraging known stable frameworks. In particular, substituting some Si atoms with germanium (Ge) is compelling due to the chemical and structural compatibility of Ge with Si. Ge lies directly below Si in the periodic table (group IV), sharing the same valence electron configuration and a tendency to form four-coordinated (sp^3) networks. Its covalent radius is only slightly larger than that of Si, allowing Si-Ge substitution with minimal lattice strain. Importantly, Ge has a smaller intrinsic band gap (Ge in the diamond structure has a 0.66 eV indirect gap and a ~ 0.8 eV direct gap at Γ) [35], so incorporating Ge into a Si lattice can narrow the gap and potentially shift the band structure toward direct-gap transitions. A recent study explored direct band gap Ge allotropes, showing that

* Corresponding author.

E-mail address: li.zhu@rutgers.edu (L. Zhu).<https://doi.org/10.1016/j.commt.2025.100045>

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hexagonal and clathrate-like phases can exhibit direct-gap behavior, which motivates further investigation of Si–Ge alloys for optoelectronic applications [36]. Continuous solid solutions $\text{Si}_{1-x}\text{Ge}_x$ are known to be feasible across the full composition range. Recent experiments have indeed demonstrated direct-gap luminescence in hexagonal SiGe alloys, with tunable emission wavelength depending on composition [37]. This finding highlights the technological appeal of hexagonal-phase SiGe: a Si-based material capable of direct optical transitions opens the door to more efficient light absorbers and emitters fully compatible with existing Si technology.

Beyond Ge, other dopants have been explored as well. For example, alloying Si with III–V compounds like AlP was predicted to produce a direct gap (~ 2.0 eV in Si_3AlP) [38]; however, that gap is too large for single-junction PV and thus suboptimal [39]. By contrast, Ge substitution is expected to lower the gap energy into a more favorable range. Alloying can also break symmetry in otherwise indirect-gap structures, converting them into direct-gap semiconductors. The key scientific question is whether we can identify Si-based alloys that achieve a direct band gap in the ~ 1.1 – 1.4 eV range (or at least improve absorption in the visible spectrum) without sacrificing stability. Successfully discovering such materials would be a significant advance: it could combine the industrial advantages of Si with optoelectronic performance on par with III–V semiconductors, potentially revolutionizing photovoltaic technology.

However, searching the vast configuration space of Si alloys is challenging. Traditional crystal structure prediction (CSP) methods are computationally expensive when exploring substitutions on large unit cells. A more targeted and cost-effective strategy is to start from known viable Si frameworks (like 4H–Si or Si_{24}) and systematically substitute Si with dopants. This still yields a combinatorially large number of possible configurations as the dopant concentration and arrangement can vary. To accelerate the discovery, we leverage high-throughput computation and machine learning (ML) to screen candidates efficiently. High-throughput DFT has previously been used to scan thousands of structures for desirable bandgaps [40–44], and recent ML models can predict materials band gaps with near-DFT accuracy much faster than direct DFT calculations [45–47]. In particular, graph-based neural networks have emerged as powerful predictors of materials properties [47–51]. Early examples like the Crystal Graph Convolutional Neural Network (CGCNN) [48] learned properties from crystal connectivity alone, and subsequent models such as MEGNet [49] and ALIGNN [50] incorporated global state parameters to improve accuracy across both molecules and crystals. These models achieve impressive performance on average, but most standard GNN architectures do not explicitly account for many-body or long-range electronic interactions [52], which are crucial for accurately predicting semiconductor band structures. As a result, generic graph models often struggle to reproduce precise band gap values or distinguishing direct vs. indirect gap character, especially in materials where subtle orbital interactions determine the band edges.

To address these limitations, we have developed EOSnet (Embedded Overlap Structure network) [53], a novel ML model that enriches the crystal graph representation with many-body orbital overlap information. In this work, we use EOSnet to prescreen Si–Ge alloy candidates for desirable electronic properties before investing computational effort in high-level DFT calculations. By integrating machine learning with density functional theory in a hierarchical workflow, we dramatically reduce the cost of discovering new materials. The scientific goal of this study is to identify viable $\text{Si}_{1-x}\text{Ge}_x$ alloy structures (derived from 4H–Si and Si_{24} prototypes) that exhibit direct or quasi-direct band gaps and enhanced optical absorption in the visible range.

2. Method

Alloy Generation and Structural Filtering: We began by enumerating possible Ge substitution patterns on two host Si allotropes: the hexagonal 4H–Si structure and the orthorhombic Si_{24} clathrate structure.

The choice of these prototypes is motivated by their intermediate band gaps and proven recoverability at ambient conditions. For 4H–Si, we used a $2 \times 1 \times 1$ supercell (8 atoms in total) to allow a range of Si/Ge ratios while maintaining the hexagonal symmetry. For Si_{24} , we used the 12-atom primitive cell for the sake of simplicity. In each case, we systematically generated $\text{Si}_{1-x}\text{Ge}_x$ alloys by substituting Si atoms with Ge in all distinct combinations (excluding the trivial $x = 0$ and $x = 1$ endpoints). The number of possible configurations grows combinatorially with cell size — e.g., a 12-atom cell has

$$\binom{12}{1} + \binom{12}{2} + \dots + \binom{12}{11} = 2^{12} - 2$$

possible substitution sets in total for various x . To manage this explosion, we applied a priori symmetry and similarity filters. We utilized the AFLOW XtalFinder routine [43,54] to eliminate duplicate structures related by symmetry or simple cell permutations. We further cross-validated AFLOW similarity ranking with an in-house fingerprint distance metric (based on local atomic environments) [55] and found excellent agreement. This procedure yielded a diverse set of candidate Si–Ge structures across a range of compositions. Quantitatively, the initial enumeration produced 254 (4H–Si) and 4094 (*Cmcm*)– Si_{24} substitution patterns; symmetry filtering reduced these to 74 and 1096 unique configurations, respectively.

Machine Learning Screening with EOSnet: Even after reducing redundancy, the pool of candidate structures remained large (on the order of thousands of unique doped configurations). To rapidly screen their electronic properties, we employed our EOSnet machine learning model [53]. EOSnet is a graph neural network that operates on crystals: each atomic site is a node enriched with a Gaussian Overlap Matrix (GOM) fingerprint [55] capturing many-body interactions in its local environment, and bonds (edges) carry interatomic distances as in standard crystal graphs. By embedding the overlap of atomic orbital Gaussian functions in the node features, EOSnet encodes higher-order angular and long-range effects that conventional message-passing networks (like CGCNN [48] or MEGNet [49]) might miss. This approach yields a rotationally invariant descriptor of the crystal that reflects not just which atoms are neighbors, but how their orbitals overlap — a factor known to influence band structure (for example, conduction-band dispersion in semiconductors depends on p – p orbital overlap across bonds). In practice, EOSnet was pretrained on a broad materials dataset to predict band gaps and other properties. We fine-tuned and evaluated the model on the specific family of Si–Ge alloys in this study, using a subset of DFT-computed band gaps as the training set. The model achieved near-DFT accuracy in predicting electronic band gaps for these alloys: for instance, on mixed 4H–Si and Si_{24} structures, EOSnet predictions have a mean absolute error (MAE) of only ~ 0.02 eV compared to DFT values (tested on a few hundreds structures held out from training). This level of accuracy is remarkable — by comparison, typical graph neural nets without explicit many-body features often struggle to reach < 0.1 eV error for band gaps. We thus proceeded to use EOSnet as a filter: any alloy configuration predicted to be metallic or having a very small gap (we chose a conservative threshold of $E_{\text{gap}} < 0.1$ eV at the PBE level) was screened out and not pursued further. This removed many candidates that would be unsuitable for photovoltaic applications.

First-Principles Calculations: After ML screening, the remaining candidates were subjected to rigorous first-principles calculations to determine their accurate band structures and optical properties. All density functional theory calculations were carried out using the Vienna ab initio simulation package (VASP). [56] We performed geometry relaxations (lattice constants and atomic positions) and band structure calculations first with the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation [57,58]. The geometry relaxations were performed using the Reformpy code, which accelerates structural optimization by mapping configurations to a fingerprint space that captures

essential structural features [59]. PBE tends to underestimate band gaps, but it reliably predicts whether a band gap is direct or indirect. The electron-ion interaction was depicted by projector augmented wave (PAW) potentials with $3s^23p^2$ and $4s^24p^2$ electrons as valence for Si and Ge atoms, respectively. We used plane-wave basis sets with a 500 eV kinetic energy cutoff and Γ -centered Monkhorst-Pack k -point meshes [60] with $0.04 \times 2 \pi \text{\AA}^{-1}$ spacing to sample the Brillouin zone.

Next, we refined the band gap values using the Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional. [61] HSE06 improves semiconductor band gaps by mixing a portion of exact exchange to mitigate self-interaction and delocalization errors in Kohn-Sham eigenvalues; it remains a ground-state functional. Excited-state and excitonic effects are treated with GW and the Bethe-Salpeter Equation (BSE). We applied HSE06 to only the promising candidates (identified as having direct/quasi-direct gaps by PBE) to obtain a reliable value of the gap in each case. For these HSE runs, we maintained the same relaxed geometries and k -point meshes.

Finally, for the most promising materials, we evaluated their optical absorption spectra. We computed the frequency-dependent dielectric function $\epsilon(\omega)$ using the GW plus Bethe-Salpeter Equation (BSE) approach [62,63], which accounts for quasiparticle corrections and excitonic effects beyond plain DFT. In practice, we performed one-shot G_0W_0 calculations [62,63] on top of the PBE orbitals to correct the band energies, then solved the BSE for electron-hole excitations as implemented in VASP. This yields the absorption spectrum (imaginary part of ϵ) including exciton effects, which is crucial for assessing photovoltaic performance. The optical calculations were computationally intensive, so we limited them to a subset of top candidates. VASPKIT utility [64] was used to generate high-symmetry k -paths for band structure plots and to post-process the optical spectra.

3. Results and discussion

The EOSnet model proved to be highly effective in predicting the electronic band gap across various elemental compositions, cell sizes, and crystal systems. We used a pretrained model (trained on 19,393 band-gap entries from the Materials Project 2024 database in our original EOSnet paper) and then fine-tuned it for the 4HSi-derived and the $Cmcm$ - Si_{24} alloys. EOSnet achieved a mean absolute error of ~ 0.02 eV in band-gap prediction on the test set (with training errors on the order of 0.011–0.015 eV). This is illustrated by parity plots of predicted vs. DFT-calculated band gaps (Fig. 1a,b), indicating our model can predict band gaps approaching DFT accuracy. Applying the fine-tuned model with a screening window of $0.2 \text{ eV} \leq E_g \leq 2.0 \text{ eV}$ further pruned the sets to 51 (4H-Si-derived) and 671 (Si_{24} -derived) structures, which we then evaluated with PBE. A similar fine-tuning and screening was also performed on larger silicon clathrate frameworks (Type-I Si_{46} and Type-II Si_{34} clathrates with Ge substitution, where Si_{46} and Si_{34} denote primitive-cell sizes; the conventional Type-II cell contains 136 atoms), and the results are shown in Fig. 1c,d. Additional training results and dataset size information are included in the Supporting Information.

These results validate that EOSnet can serve as a reliable surrogate for DFT in this materials family. Practically, the ML screening pruned the candidates drastically: out of the hundreds of unique substitution patterns generated, a large fraction were predicted to be either metallic or to have very low/high gaps and were discarded. This left 51 (4H-Si-derived) and 671 (Si_{24} -derived) unique structures to be examined with DFT, a huge reduction in computational workload. Notably, EOSnet identified all the eventual winners (direct-gap candidates) correctly as semiconducting with moderate gaps, ensuring that none of the promising structures were mistakenly filtered out.

After performing PBE calculations on the ML-selected set, we found 19 distinct Si-Ge alloy structures that exhibit a direct or quasi-direct band gap. Fifteen of these originated from the 4H-Si prototype and four from the Si_{24} prototype. A quasi-direct (QD) gap here refers to cases where the fundamental band gap is formally indirect but there is a

Table 1

Calculated band gap E_g (eV) with PBE and HSE06 hybrid functional for selected Si-Ge alloys. QD, and D represent quasi-direct, and direct band gaps, respectively.

Crystal	E_g		Type
	PBE	HSE	
$P6_3mc$ - Si_2Ge_6	0.0147	0.6565	D
$P3m1$ - Si_3Ge_5	0.2474	0.8781	QD
$P6_3mc$ - Si_4Ge_4	0.1681	0.8285	D
$P3m1$ - Si_4Ge_4	0.3570	0.9673	QD
Cm - Si_5Ge_7	0.0557	0.5043	QD
Cm - Si_6Ge_6	0.1641	0.6936	D

direct transition very close in energy (within a few tens of meV), meaning the material would behave nearly like a direct-gap semiconductor for optical purposes. For clarity, Table 1 summarizes a representative subset of six top candidates, listing their PBE band gap (E_g^{PBE}), HSE06 band gap (E_g^{HSE}), and whether the gap is direct (D) or quasi-direct (QD).

All six examples above have their HSE06 gaps in the 0.5–1.0 eV range. In all cases, HSE yields a larger gap than PBE (as expected for semiconductors), typically by 0.3–0.6 eV. The maximum HSE gap we found among any of the 19 candidates was about 0.97 eV (for the $P3m1$ - Si_4Ge_4 listed). While these values fall short of the ideal 1.1–1.4 eV range for single-junction solar absorbers, they are substantially closer to that target than the gaps of the parent phases or pure Ge. For example, pure 4H-Si has an indirect gap 0.8–0.9 eV and $Cmcm$ - Si_{24} has a quasi-direct gap ~ 1.3 eV (PBE values) but with low optical activity near the edge, whereas introducing Ge yields direct-gap behavior albeit at somewhat lower gap energies. We emphasize that direct gap character was achieved in multiple instances: e.g. $P6_3mc$ - Si_2Ge_6 , $P6_3mc$ - Si_4Ge_4 , and Cm - Si_6Ge_6 are direct-gap semiconductors by HSE06 (with conduction band minimum and valence band maximum at the same k -point in the Brillouin zone). The others are quasi-direct, often with the conduction band minimum shifted slightly off the zone center (for instance, in the Cm structures derived from Si_{24} , the CBM occurs at an M -point very close in energy to the Γ -point valence band maximum).

To visualize the electronic structure, Fig. 2 shows the calculated HSE06 band structures for the above six candidates. In the four hexagonal-derived alloys (Fig. 2a–d), the band gap is located at the Brillouin zone center Γ . In the two orthorhombic/monoclinic alloys (Fig. 2e,f), the gap occurs at a zone-boundary point labeled M (specifically M2). We find that Ge substitution tends to raise the valence band maximum (making it less negative with respect to the vacuum level) and often lowers the conduction band minimum, relative to the parent Si frameworks, which is consistent with the smaller gap of Ge and its contribution to bonding/antibonding levels. Notably, the valence-band dispersion near the top is not flattered by the presence of Ge; instead, the valence bands remain quite dispersive (especially in the direct-gap cases), which is a positive sign for carrier mobility. We also notice that all these alloys show reasonably steep band edges at both VBM and CBM, indicating relatively small effective masses for holes and electrons. This suggests that carriers in these materials should retain good mobility despite the modifications to the band structure.

Ge incorporation was found to systematically lower the band gap of the host Si structures, which is expected due to the smaller band gap of Ge. More interestingly, certain ordered arrangements of Ge atoms created conditions for direct transitions. In hexagonal 4H-derived alloys, we find that having Ge occupy specific lattice sites (resulting in space groups $P6_3mc$ or $P3m1$) can break inversion symmetry and alter the conduction band such that the CBM moves to Γ . The direct-gap alloys from the 4H family typically had the highest Ge content (50% or more Ge), suggesting that a high concentration of Ge is helpful in pushing the material toward direct-gap behavior. This makes sense: pure hexagonal Ge is a direct-gap semiconductor (with $E_g \approx 0.3$ eV in

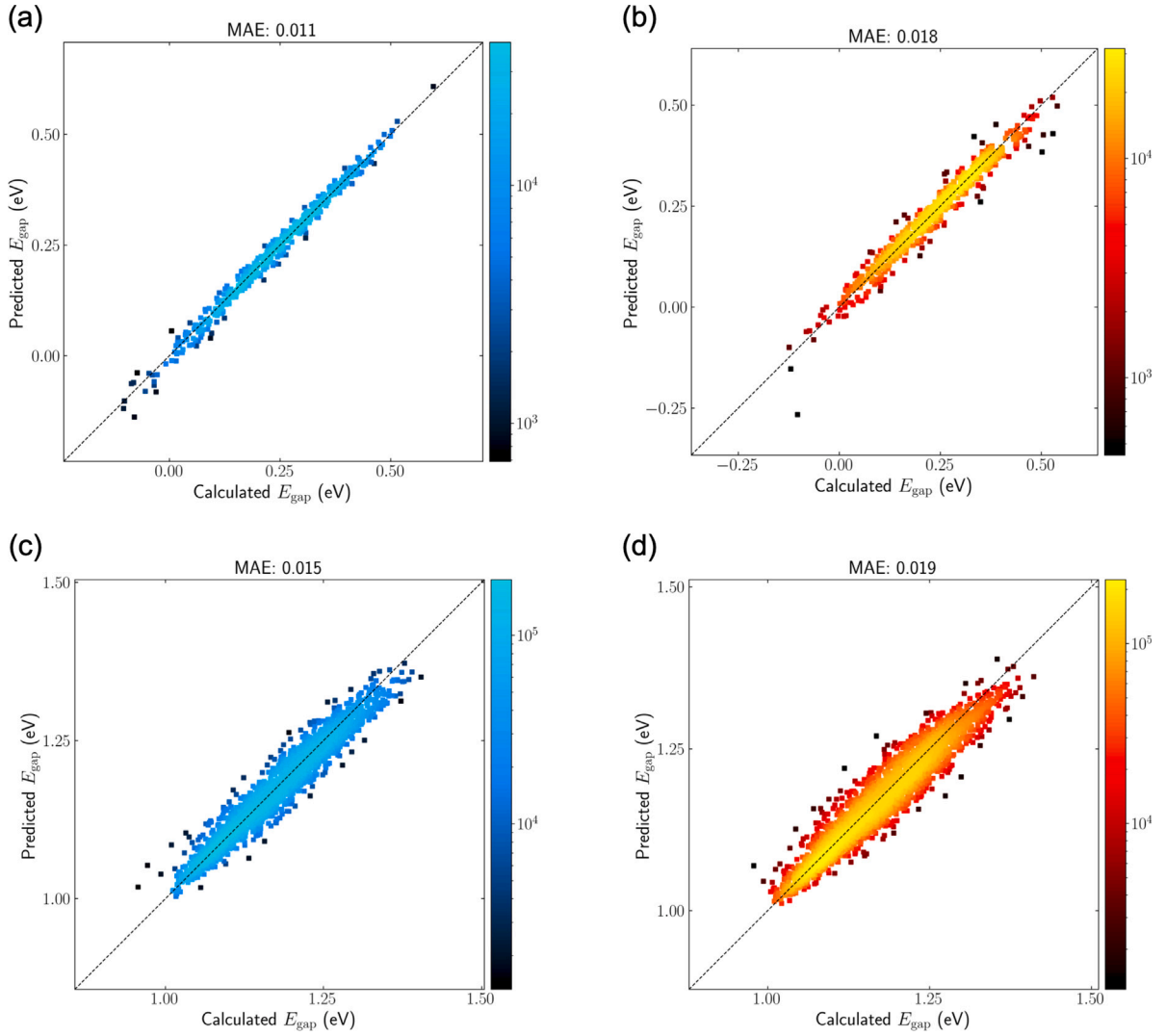


Fig. 1. Top panel: Parity plots of predicting the electronic band gap (E_{gap}) for germanium substituted 4H-Si and C_{mcm} -Si₂₄ prototypes with a MAE of (a) 0.011 eV of 468 structures for training and (b) 0.018 eV of 585 structures for prediction. Bottom panel: Parity plots for the prediction of electronic band gap (E_{gap}) for germanium substituted type-I (Si₄₆) and type-II (Si₃₄) silicon clathrate prototypes with a MAE of (c) 0.015 eV of 1982 structures for training and (d) 0.019 eV of 2478 structures for prediction.

the hexagonal form), whereas pure hexagonal Si (4H-Si) is an indirect-gap semiconductor with a larger gap. By mixing the two, one can interpolate between these electronic structures. Our findings imply a trade-off: higher Ge fractions favor direct gaps but also yield smaller gap energies. Indeed, none of our best candidates reached the 1.1 eV threshold, likely because approaching that optimal gap would require a lower Ge fraction at which the band structure reverts to indirect (like 4H-Si). This mirrors experimental observations in hexagonal SiGe alloys, where compositions around Si_{0.2}Ge_{0.8} emitted light at ~0.8 eV (direct gap photoluminescence), and adding more Si increased the emission energy but eventually lost the direct nature of the gap. In our study, $P3m1$ -Si₄Ge₄ (50% Ge) had the largest gap (~0.97 eV HSE) among the direct/quasi-direct set, but it was quasi-direct; a bit less Ge might push it above 1 eV but then the CBM might shift off Γ . This highlights a fundamental compromise in alloy design: achieving the right gap energy and direct gap simultaneously is challenging when the alloy end-members have disparate electronic characteristics.

We calculated the imaginary part of dielectric function by solving the BSE for several of the direct-gap candidates to quantify their ability to absorb visible light. Fig. 3 presents the results for six representative alloys (the same set as in Table 1) in terms of the imaginary part of

the dielectric function $\epsilon_2(\hbar\omega)$. For comparison, the spectra of d -Si (diamond silicon) and the original 4H-Si and Si₂₄ structures are also shown. The vertical dashed line in the figure indicates the approximate lower energy bound of the visible spectrum (around 1.7 eV, 730 nm). All the Si-Ge alloys exhibit markedly stronger absorption in the visible range than d -Si or their undoped counterparts. This is a direct consequence of the improved band gap directness and the involvement of Ge-related electronic states that increase the transition dipole matrix elements.

Across all the alloy spectra (violet curves in each sub-figure of Fig. 3), absorption onsets at lower photon energy and with higher magnitude than in pure silicon. For instance, diamond Si has virtually no absorption until photon energy exceeds 1.5–2 eV (due to its indirect gap), whereas the Ge-alloyed structures start absorbing from around 0.5–0.8 eV upward (their direct or quasi-direct gap energies), and their absorption in the visible light range is significantly higher. Even though none of these alloys has an ideal 1.1–1.4 eV direct gap, their stronger absorption in the visible indicates they can more efficiently utilize sunlight within the device's spectral response range. For example, $P6_3mc$ -Si₂Ge₆ (75% Ge) and $P6_3mc$ -Si₄Ge₄ (50% Ge) both show continuous absorption through the entire visible range, outperforming 4H-Si (which itself outperforms d -Si). This can be attributed to the allowed direct transitions at lower energies introduced by Ge. We also

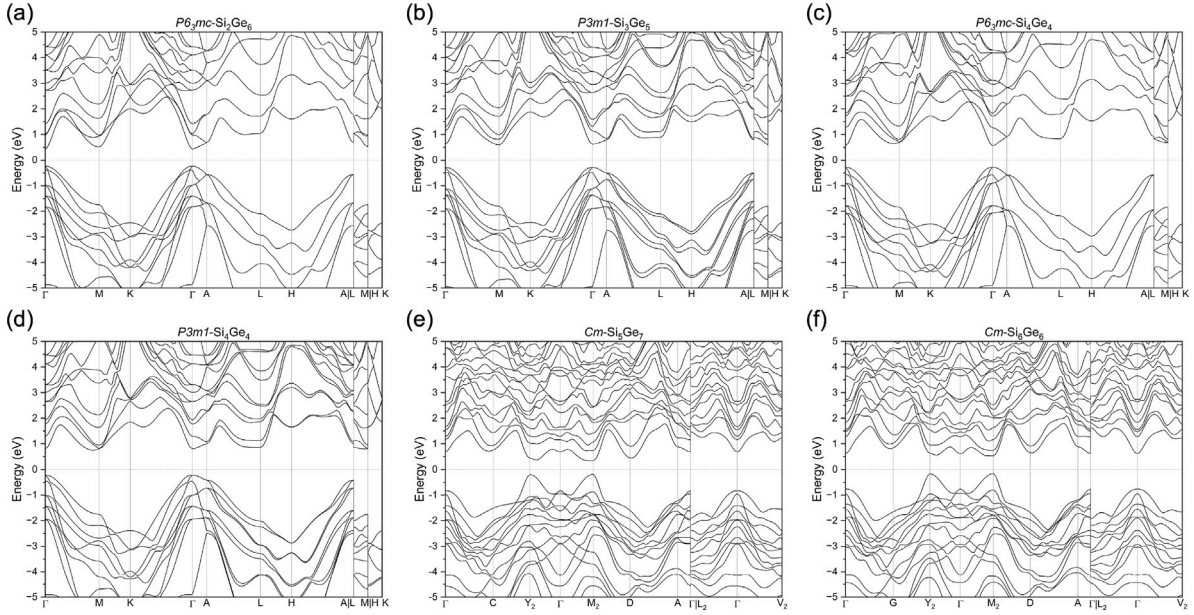


Fig. 2. Calculated electronic band structures using the HSE06 hybrid functional at ambient pressure with the Fermi level set to zero. Band gap locations are at Γ points for $P6_3mc$ - Si_2Ge_6 , $P3m1$ - Si_3Ge_5 , $P6_3mc$ - Si_4Ge_4 and $P3m1$ - Si_4Ge_4 ; band gap locations are at M_2 points for Cm - Si_5Ge_7 and Cm - Si_6Ge_6 .

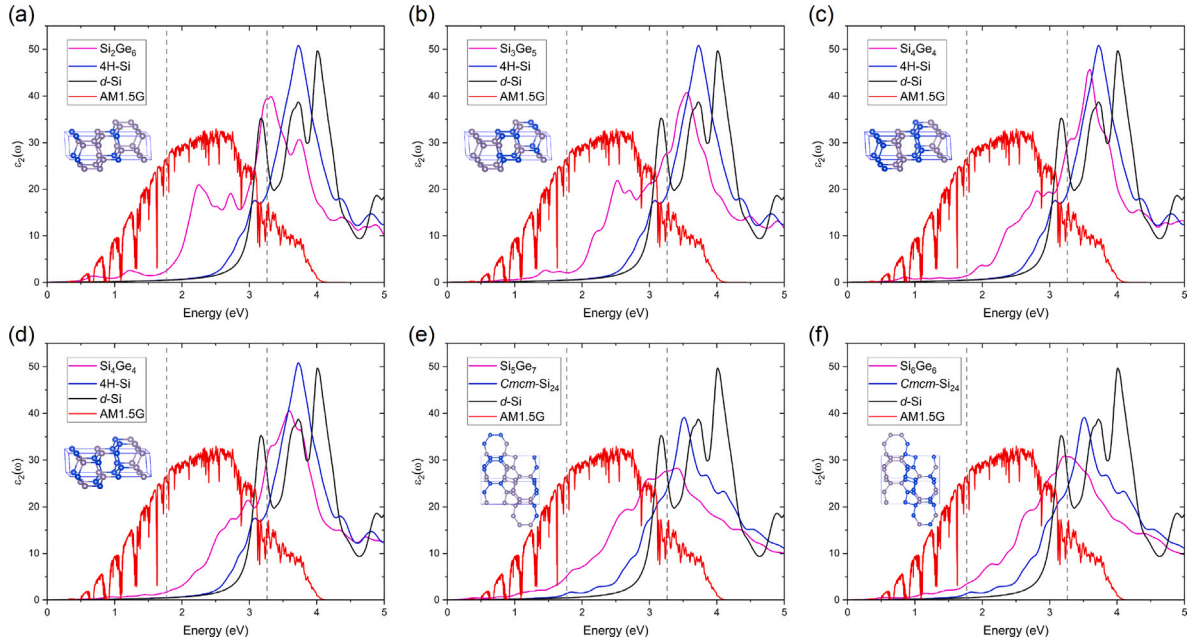


Fig. 3. Calculated imaginary part of dielectric function (ϵ_2) and top views of the crystal structures for (a) $P6_3mc$ - Si_2Ge_6 , (b) $P3m1$ - Si_3Ge_5 , (c) $P6_3mc$ - Si_4Ge_4 , (d) $P3m1$ - Si_4Ge_4 , (e) Cm - Si_5Ge_7 and (f) Cm - Si_6Ge_6 . Blue and grayish lavender spheres represent Si and Ge atoms, respectively. The blue solid lines denote the boundary of the unit cells. Red curve is the Air Mass 1.5 Global tilted (AM1.5G) solar spectral irradiance which is obtained from <https://www.nrel.gov/grid/solar-resource/spectra-am1.5>.

note that the absorption spectra of the alloys exhibit prominent peaks in the visible, corresponding to excitonic transitions. The presence of Ge tends to increase the density of states near the band edges, which in turn enhances the magnitude of ϵ_2 . Additionally, the band curvature at the band edges, as mentioned, is not too flat, which is beneficial because overly flat band edges (large effective mass) would reduce absorption near the onset (due to fewer available states for transitions). Overall, Ge substitution clearly boosts the optical absorption of silicon allotropes in the desired wavelength range, a positive indication for photovoltaic applications. While the absolute band gap positions are somewhat below optimal, the quality of the band gap (direct with

strong oscillator strength) is significantly improved in these Si-Ge alloys. In terms of applications, these properties suggest suitability for thin absorbers and as bottom cells in tandem architectures where sub-eV to 1 eV gaps can be advantageous for current matching. To quantify device-relevant trends, we also computed a thickness-normalized upper bound to the short-circuit current from the GW-BSE absorption (thin-film limit); the Si-Ge alloys exceed d -Si and their undoped hosts by roughly a factor of 2–3 (see Table S4 in the Supporting Information).

Phonon dispersions for the six highlighted alloys show no imaginary branches at ambient pressure (see Supporting Information, Fig. S3), indicating dynamical stability. Calculated formation enthalpies at 0 K

lie in the 20–31 meV/atom range (Table S3), comparable to values reported for recoverable Si frameworks (e.g., clathrates). While positive at 0 K, these magnitudes are consistent with synthesis via templating or epitaxial routes and may be offset by configurational/vibrational entropy at growth temperatures, supporting plausible synthesizability within the recovered 4H/Si₂₄ backbones.

4. Conclusion

In summary, this work demonstrates a combined machine learning and first-principles approach for the accelerated discovery of Si-based photovoltaic materials. By systematically investigating germanium-doped silicon allotropes derived from 4H-Si and *Cmcm*-Si₂₄ prototypes, we identified several promising Si_{1-x}Ge_x alloys with direct or quasi-direct band gaps and enhanced optical absorption in the visible range. Our EOSnet machine learning model, which incorporates many-body orbital overlap information, enabled rapid and accurate screening of alloy candidates, reducing the computational cost of high-level DFT calculations. The model achieved near-DFT accuracy in band gap prediction, ensuring reliable identification of semiconducting and direct-gap structures. First-principles calculations confirmed that Ge substitution can induce direct-gap behavior and significantly improve light absorption compared to pure silicon allotropes. Although the absolute band gap values remain slightly below the optimal range for single-junction solar cells, the improved directness and absorption strength highlight the potential of Si-Ge alloys for next-generation photovoltaic applications. This hierarchical workflow combining machine learning screening with targeted DFT and GW-BSE optical calculations provides an efficient strategy for discovering new functional materials. The approach can be extended to other alloy systems and crystal prototypes, paving the way for rational design of advanced optoelectronic materials.

Despite these successes, the alloys discovered have band gap values that are slightly below the optimal window for single-junction solar cells. None of the candidates yet reaches the ~1.1 eV direct gap ideal. Additionally, our study focused on ordered configurations; real materials may contain disorder or require non-equilibrium fabrication, which could affect their electronic properties. The stability of high-Ge-content hexagonal SiGe at operating conditions is not yet known. From a modeling perspective, the ML model was applied to a specific chemical/structural space and would require retraining or transfer learning to handle vastly different materials (though the EOSnet approach is general in principle). While challenges remain in achieving the perfect band gap and in material fabrication, the insights gained — particularly the efficacy of Ge substitution and the use of machine learning in materials discovery — provide a clear direction for future research in the design of high-performance semiconductors for solar energy conversion.

CRediT authorship contribution statement

Shuo Tao: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Li Zhu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.commt.2025.100045>.

Data availability

Data will be made available on request.

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